

1.2 Register Data

We use several administrative registers at Statistics Denmark. Our matched employer-employee data come from the *Employment Statistics of Employees* (BFL), which records earnings, hours, occupation, and industry for all job spells in Denmark on a monthly basis from 2008 onward. This register is compiled by the Danish tax authorities and subsequently harmonized by Statistics Denmark. We complement this with demographics data on individuals from the *Population Register* (BEF), and wealth information from the *Personal Wealth Register* (FORMPERS). Finally, we draw firm financial data (e.g., annual revenue and employment) from the *Firm Statistics* (FIRM) Register. Because our survey was sent to workers identified in the register data by their (deidentified) social security numbers (*pnr*), all respondents can be matched to the register data. While labor market data in the BFL register are updated continuously, we only have firm financial data up to 2022. For that reason, we focus our effect analyses on labor market outcomes.

1.3 Survey Sample

We invited 115,000 workers to participate in each of our survey rounds in 2023 and 2024. The registers at Statistics Denmark enabled us to target these invitations by occupation and workplace. In particular, for each of our 11 occupations, we conducted a workplace-based sampling procedure, first drawing a random set of workplaces within each occupation and then sampling all relevant workers at these workplaces. This sampling procedure maximizes the statistical power of the workplace-level analyses while keeping our sample representative. Appendix B.1 details the sampling protocol. We sent three reminders per survey round, two by e-mail and one by text. The invitation letters are in Appendix I. We received about 25,000 valid and complete responses to each survey. Appendix B.2 details our survey response rates. While our main analysis focuses on responses from the 2024 round, we use the 2023 round to examine early adopters.

1.3.1 Representativeness and Response Quality

Appendix B.2.1 conducts several checks on the representativeness and quality of our survey responses. These analyses extend the checks in Humlum and Vestergaard (2025) to the 2024 survey round. First, we ensure that our sample represents the population based on observables, including age, gender, experience, earnings, and wealth. Second, following Dutz et al. (2025), we use randomized participation incentives to show that our findings are also balanced on workers’ latent willingness to participate in the survey. Finally, we cross-check that the survey responses align with variables that are also recorded in the administrative registers.

1.4 The Landscape of AI Chatbot Adoption

By the end of 2024, employers and employees have taken clear positions on AI chatbots: adoption is widespread across all 11 exposed occupations, but strategies vary markedly within each. Appendix D provides detailed evidence on adoption and workers’ reported experiences, and Appendix A.1 develops a Roy-style adoption model to interpret these patterns. Here, we highlight the key points that set the stage for our analysis of labor market effects.

Most employers now encourage chatbot use. As Figure 1 shows, 43% of workers are explicitly encouraged to use them, another 21% are allowed, and only 6% are prohibited. Employers also make tangible investments in chatbots: among encouraged workers, 39% have received training, 61% have access to an enterprise chatbot, and 29% have both.⁵

Many otherwise similar firms have adopted markedly different strategies. Observable firm characteristics (age, size, productivity, and ownership) explain only 1.3% of the within-occupation variation in encouragement—well below the 10.6% explained by occupation alone (Appendix D.1).

Encouragement is associated with substantially higher adoption and reported benefits. Even without employer initiatives, 40% of workers have used chatbots at work and 6%

⁵Among non-encouraged workers, 21% have access, 11% have training, and 4% have both.

use them daily. Despite this high baseline, Table 1 shows that encouragement raises these rates by 35 and 11 percentage points, respectively (Columns 1-2), and is consistently associated with higher reported time savings, improved quality, and enhanced creativity (Columns 3-6).⁶ By contrast, training and enterprise tools alone do little to increase daily use and are associated with smaller reported gains.⁷ For this reason, our subsequent analysis uses encouragement as the primary workplace treatment.

AI chatbots are reshaping the nature of work. Among chatbot users with employer initiatives, about 12% report new tasks from chatbots (Table 1, Column 8). These tasks include both productive activities (e.g., content drafting, ideation) and non-productive ones (e.g., integration, compliance, oversight). Appendix B.3 classifies this new work using workers’ free-text descriptions. New tasks extend even to non-users (Column 10), signaling broader workplace transformations.

Substantial variation in use and perceived effects persists even among similar workers facing identical employer policies. While reported productivity gains are modest on average (2.8% time savings for the typical user; Appendix D.3), some report saving more than an hour per day, alongside improvements in quality, creativity, and new tasks. Section 3 exploits this variation to test whether labor market impacts differ for those reporting larger effects.

1.4.1 Robustness of Employer Initiative Measures

Before using our measures of employer initiatives in the main analysis, we assess their validity with a series of robustness checks.

First, Appendix E.4 shows that all correlations with employer initiatives in Table 1 remain stable when controlling for firms’ characteristics and workers’ detailed task mixes within occupations. This ensures that differences across initiatives are not driven by firm age, size, or productivity, nor by variation in the types of tasks more amenable to

⁶Appendix A.1 presents a Roy-style model of chatbot adoption to help interpret the estimates in Table 1 on how employer initiatives affect adoption and user-reported benefits.

⁷As Appendix D discusses, training and enterprise tools for non-encouraged workers are likely designed to prevent misuse rather than to enhance productivity.

chatbot use. In addition, the panel analyses in Section 3 confirm that adopters of different initiatives followed similar labor market trajectories prior to the arrival of AI chatbots.

Second, because our data come from workers rather than senior leadership, our estimates reflect initiatives *as perceived by workers*. Policies may not be perfectly communicated, and self-reports may diverge from employer intentions. Nonetheless, Appendix E.5 shows that our results are robust when employer initiatives are measured using coworker reports, indicating that these measures capture workplace-wide initiatives.

2 Empirical Strategy

Evaluating the labor market impacts of AI chatbots raises several conceptual questions: What are the relevant treatments to consider? If the treatments have an effect, in which labor market outcomes would they manifest? And how can we identify these impacts in the data?

Motivated by the dual nature of adoption—with variation across workplaces depending on their chatbot initiatives and within workplaces depending on individual use—we analyze impacts at both the worker and workplace levels. Appendix A.2 presents a theoretical framework for how adoption translates into labor market outcomes, guiding the following analyses:

1. *Worker earnings:* If AI chatbots raise individual productivity, these gains may translate into higher wages for users (Krueger, 1993). Notably, the wage effects depend on whether adoption is worker- or firm-driven, the productivity and amenity value of chatbot use, and the relative scarcity of adopting workers versus firms. Section 3.1 investigates these predictions by estimating the impacts of chatbot adoption on earnings by workers’ intensity of use, employer initiatives, reported productivity gains, and new AI tasks.
2. *Workplace employment:* If chatbots reduce the need for labor, adopting workplaces might slow hiring or lay off workers. Conversely, if the tools boost productivity and